



Demand shifting bids in energy auction with non-convexities and transmission constraints



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ARTICLE INFO

Article history:

Received 30 September 2013

Received in revised form 13 March 2015

Accepted 19 May 2015

Available online 18 June 2015

Jel classification:

C61

D24

D44

D51

D63

Q41

Keywords:

Double auction

Unit commitment

Energy pricing

Minimum uplift

Individual maximum profits

ABSTRACT

The major objective of this paper was to propose clearing and pricing models suitable for demand shifting bids in the efficient, but non-convex pool-based auction. Complex generators' offers bring non-convexities into the efficient auctions due to e.g. start-up costs and times. This paper focused on the responsive demands, introducing simple, yet adequate linear constraints into a multi-period bid/offer-based optimal power flow (OPF DC) model. As the standard locational marginal prices (LMPs) may not support the auction outcomes due to non-convexities, uplifts are needed to reduce generators' loss. Previous work has developed a minimum-uplift pricing model that directly optimizes prices, so that uplifts arising from generators' profit-suboptimality and simple, elastic demands' benefit-suboptimality are minimized. This work extended the mixed integer linear programming (MILP) formulation of the previous model to incorporate new linear constraints defining benefit-suboptimality of demand shifting bids. Furthermore, the transmission constrained market was attempted. As a result, the buyers were protected against over-curtailment; moreover, prices complemented with minimum uplifts were fair for both generators and demands. The models were validated on the literature-based cases, including IEEE RTS 24-node 24-hour system.

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1. Introduction

The role of active demand is now widely recognized in the energy market. Among others, an active demand-side participation may reduce the effect of the generation and transmission capacity constraints, price spikes, fuel consumption and market power exercises (e.g., see Fang et al. (2012); Khodaei et al. (2011); Su & Kirschen (2009)). To encourage the active demand participation new energy trading mechanisms are designed and developed, that incorporate certain types of demand response programs (Sioshansi, 2008). Most close to pure market competition is demand bidding.

Consumers submitting bids in energy auctions have to be flexible with the timing of consumption. Even though there are several buyers, who can meet this requirement (large industrial consumers, on-site storage, distributed generation technologies, etc. – see Zoltowska (2013)), some restrictions on the flexibility may be desired to protect the customers against over-curtailment (Borghetti et al., 2001). This paper therefore proposes additional requirements submitted together with elastic demand bids.

A centralized pool-based auction is an example of a comprehensive trading mechanism that can benefit the most from the elastic demand

bids (Singh et al., 2011). The possible demand response directly affects the generating units' schedules, which are the outcomes of pool-based auction clearing. The market operator (e.g. US ISO) solves the competitive version of optimal power flow (OPF) problem, based on the multi-period offers submitted by generators (Stoft, 2002).

Offers bring non-convexities into the auction due to e.g. start-up costs and times, which are known to cause pricing problems, since the Madrigal & Quintana (2001) preliminary work.

A common practice is to use a locational marginal pricing (LMP) scheme, which ignores non-convex costs components. Additional payments, known as uplifts, are necessary to recover these offer-based costs. Uplifts may yield incentive and equity issues, especially in a transmission congested system, where high additional payments may be required to restore uncovered costs. Moreover, opportunity costs may appear. Thus, the natural effort of research is to minimize the impact of uplifts (Litvinov, 2010).

1.1. Literature review

There are many papers that deal with demand response, focusing on specific programs that analyze the economic models or the program's effects on peak load reductions, etc. (e.g., Kim & Shcherbakova, 2011). Recently, also double pool-based auctions are studied more widely,

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and several optimization models are developed (Fang et al., 2012; Motto et al., 2002).

David & Li (1993) for the first time recognized the demand dependency on prices at several time intervals. They point out that customer load will migrate from one time period to another if there is a commensurate price attraction. A paper by Su & Kirschen (2009) introduced a day-ahead market-clearing mechanism that allows price-sensitive demand-side bids with load shifting. Singh et al. (2011) further utilize this model for congestion management. Both papers (Su & Kirschen, 2009; Singh et al., 2011) consider the simplest form of demand shifting bid (no restrictions on the amount of shifting energy). The intertemporal characteristics of demand bids were incorporated into market clearing by Khodaei et al. (2011). The solution consists of iterative process between the master problem and subproblems.

All the aforementioned mechanisms assume the convex generating unit offers, thus simple marginal prices may be applied.

Recently, the impact of demand response on outcomes of pool-based auctions with non-convexities is gaining more attention. Magnago et al. (2015) point negative implications on social welfare due to load curtailment, when only simple, price-sensitive demand bids are considered. Parvania et al. (2014) provide detailed model of the complex technical characteristics of demand shifting bids, including minimum/maximum duration for daily load reduction or reduction initiation cost, to equip the ISO with a framework integrating demand response with pool-based auction. But the issue of adequate pricing is not addressed.

Literature on the pricing non-convexities is extensive, see for example O'Neill et al. (2005) or Andrianesis et al. (2013) for the recent study. Various approaches have been proposed so far, but the elastic demand typically is neglected, and transmission constraints are omitted.

One trend, followed up by the author of this paper, develops minimum-uplift pricing schemes, as for the first time introduced by Hogan & Ring (2003). This pricing mechanism is currently under intense consideration (e.g., Gribik et al., 2007; Toczyłowski & Zoltowska, 2009; Ye et al., 2014), as it aims in developing prices that minimize uplift.

The paper by Toczyłowski & Zoltowska (2009) included for the first time the simple elastic demand bids in pool-based auction pricing scheme. The scheme is based on relative buying and selling prices optimization, so they are compliant with both producers' and consumers' utility maximization goals and the difference (weighted) between the buying and selling prices is equal to minimum uplift. This approach explicitly minimizes the uplifts by considering both the uplifts and the market prices as variables, moreover it gives an answer how to share the uplift cost among profit-driven generators and consumers.

The convex hull pricing approach proposed by Gribik et al. (2007) minimizes uplifts through a complex Lagrangian objective function. The method has been extended by Ye et al. (2014) to account for flexible demand non-convexities and benefit loss.

Both approaches (Toczyłowski & Zoltowska, 2009; Ye et al., 2014) consider only network unconstrained auction. Still, it is important to further enhance the studies, as for example Wu (2013) points out significant pricing problems when demand response displaces less flexible generating units and/or causes additional transmission congestion. This paper addresses these issues.

1.2. Contribution

This paper makes a new contribution in the fields of the double pool-based auction clearing and pricing.

The formulation of demand shifting requirements in auction clearing model has an advantage over the other approaches (e.g., Khodaei et al., 2011; Wu, 2013) as the binary curtailment state of the load is not needed, still the major load operation constraints are maintained.

Also, this paper formulates for the first time linear constraints that define benefit-suboptimality of demand shifting bids. These constraints are included in the direct minimum-uplift model (DMU) developed in the previous paper by Toczyłowski & Zoltowska (2009). Moreover, the model

is enhanced to include transmission constraints, so the prices reflect not only generators' and demands' constraints, but also network congestion.

Clearing and pricing models are formulated as mixed-integer linear programming (MILP) problems, with linear constraints related to shifting demands. The resulting prices and uplifts ensure optimal revenues/-payments for all consumers and generators. The models are computationally tractable and can be solved by using standard branch-and-cut solvers.

1.3. Paper organization

The paper is organized as follows. The shifting requirements that can be simply included in the bid/offer-based unit commitment model are introduced in Section 2, while the clearing model is shown in Section 3. The corresponding pricing model that minimizes uplifts and maximizes the outcomes of all auction participants is explained in Section 4. The case studies are reported in Section 5. Then conclusions are stated in Section 6. Finally, the nomenclature is provided.

2. Demand shifting bids

Each buyer d is allowed to submit multi-part bids, that consist of:

Fixed component FD_{dh} : denoting hourly amount of power required by buyer d at any price; if the buyer is fully elastic fixed component is equal zero.

Elastic bid: typical elastic bid (e.g., Toczyłowski & Zoltowska, 2009; Motto et al., 2002) is assumed, i.e. bid specifies I^D energy blocks in a non-increasing price order. Block i defines an energy-price pair, where e_{dhi} denotes the unit price and $\bar{p}_{dhi}^D$ denotes the maximum quantity offered to buy by buyer d in hour h . Fixed demands do not submit elastic bids, in other words $\bar{p}_{dhi}^D = 0$ in each block i . Total power submitted in all blocks defines maximum hourly elastic load $p_{dh}^{max} = \sum_i \bar{p}_{dhi}^D$.

Expected elastic load $\bar{p}_{dh}$: this optional component doesn't play any role in mathematical models, but is useful to interpret the scheduled loads in terms regarding responsive demands. Expected load must be defined between zero and maximum elastic load.

Shifting constraints: in this paper we propose that the buyers are also allowed to limit the elastic daily consumption between E_d^{min} and E_d^{max} megawatthours. Typically E_d^{max} equals total maximum hourly elastic load. The following dependencies between E_d^{min} and E_d^{max} are possible:

$E_d^{min} = 0$ Fixed component is enough to meet the buyers' demand; however, in a case of low prices additional power may be bought (in the amount not exceeding E_d^{max} in general, nor p_{dh}^{max} in specifics). This kind of constraint may be convenient for example for storage units. From another point of view $E_d^{min} = 0$ indicates that the expected load may be curtailed without any consequences.

$E_d^{min} = E_d^{max}$ The limit on E_d^{min} to be equal to E_d^{max} indicates that all the load curtailed must be shifted to other hours. This kind of constraint may be convenient for example for electric vehicles (EVs), who may submit elastic bids in hours when they are available to charging.

$E_d^{min} \leq E_d^{max}$ This general relation creates various curtailment-shifting possibilities.

Fixed and elastic bid's components are illustrated in Fig. 1.

Shifting constraints are mathematically represented as:

$$E_d^{min} \leq \sum_{h,i} p_{dhi}^D \leq E_d^{max}, \quad \forall d, \quad (1)$$

where scheduled non-negative loads p_{dhi}^D must not exceed the maximum limits stated in elastic bids:

$$p_{dhi}^D \leq \bar{p}_{dhi}^D, \quad \forall d, h, i. \quad (2)$$

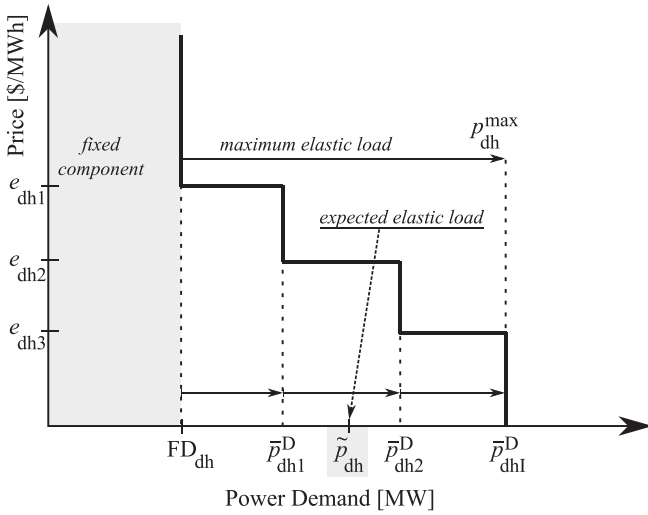


Fig. 1. Exemplary bid of buyer d in hour h .

This formulation has an advantage over the previous ones (e.g. by Wu (2013); Khodaei et al. (2011)) since the binary curtailment state of the load is not needed, still the major load operation constraints are maintained.

3. Pool-based auction clearing

The market clearing is performed by the market operator, who determines feasible and efficient hourly schedules.

Besides the demand bids there are complex generator offers. Each offer specifies set of P energy blocks in a non-decreasing price order. Block i defines an energy–price pair, where c_{lhi} denotes the unit price and $\bar{p}_{lhi}^S$ denote the maximum quantity offered to sell by generator l in hour h . The first energy block $\bar{p}_{lhi}^S$ states the minimum generation capacity of the l th generator. Furthermore, cost components UC_l (such as no load cost, and start-up) connected with commitment variables are considered for each l th generator and operating constraints $\mathcal{V}_l$.

Mathematically, the constraints on supply offers can be represented as:

$$\bar{p}_{lhi}^S v_{lh} \leq p_{lhi}^S \leq \bar{p}_{lhi}^S v_{lh}, \forall l, h, i, \quad \vec{v}_l \in \mathcal{V}_l, \quad \forall l, \quad (3)$$

where the vector of hourly commitment variables $\vec{v}_l = \{v_{l1}, \dots, v_{lH}\}$ sets the on/off statuses of the l th generator.

The market operator maximizes social welfare Q defined as a difference between consumption willingness to pay (fixed component is omitted in the objective) and the generation offered cost:

$$\max Q = \sum_{d,h,i} e_{dhi} p_{dhi}^D - \sum_l \left(\sum_{h,i} c_{lhi} p_{lhi}^S + UC_l(\vec{v}_l) \right). \quad (4)$$

Besides the constraints on the purchase bids and supply offers Eqs. (1)–(3), the power balance must be ensured:

$$\sum_n \sum_{l \in N_n^S} p_{lhi}^S = \sum_n \sum_{d \in N_n^D} \left(FD_{dh} + \sum_i p_{dhi}^D \right), \perp \chi_h, \forall h, \quad (5)$$

where χ_h is a shadow price of Eq. (5).

Shift factors SF_k are used to define load flow:

$$P_{kh} = \sum_n SF_k^n \left(\sum_{l \in N_n^S} p_{lhi}^S - \sum_{d \in N_n^D} \left(FD_{dh} + \sum_i p_{dhi}^D \right) \right), \quad \forall k, h. \quad (6)$$

Finally, the load flow and counterflow in each line are constrained with transmission capacity limits:

$$P_{kh} \leq U_k, \perp \gamma_{kh}^+, \quad -P_{kh} \leq U_k, \perp \gamma_{kh}^-, \quad \forall k, h, \quad (7)$$

where each transmission constraint (7) has its own shadow prices γ_{kh}^+ or γ_{kh}^- .

The optimal solution provides schedules to be priced. Standard method used by the ISOs derives nodal prices (LMPs) from the shadow prices associated with the constraints (5) and (7):

$$LMP_{nh} = \chi_h - \sum_k SF_k^n (\gamma_{kh}^+ - \gamma_{kh}^-), \quad \forall n, h, \quad (8)$$

out of LP solution with fixed commitment variables.

4. Prices and uplift determination

The pricing problem is a consecutive market settlement step, after the efficient schedules were determined. The clearing models allow us to maximize the social welfare under generators' and loads' operating constraints, but the fair market-clearing prices for each hour cannot be explicitly obtained. LMPs as defined in Eq. (8) do not reflect commitment costs nor constraints, resulting in uplifts. This paper formulates a new pricing model that provides prices more compliant with shifting demands' and generators' profit maximization goals, i.e. that minimize uplifts and reflect network constraints.

The model builds upon the previous co-authored papers by Toczyłowski & Zoltowska (2009) and Zoltowska & Toczyłowski (2008) in the field of individual uplifts modeling while the new model is stated. It is a nontrivial extension to previous works, as the shifting demand benefit maximization is addressed. Moreover, transmission constraints are considered.

The idea of uplift minimizing pricing scheme is as follows. Uplifts (compensations) are paid separately from market clearing price payments to unfairly priced buyers and sellers. Uplifts cover the difference between the optimal and actual benefits/profits. They are directly minimized in the model that sets out relation between uplifts (variables), prices (variables), market efficient schedules (parameters) and individual bids/offers (parameters).

4.1. Demand uplift and benefit formulation

The model of uplifts assigned to shifting demand bids is fully explained in this subsection.

The shifting demand bids involve the time-coupled constraints, thus the adequate demand pricing may not be addressed independently in particular hours, as it was done in the previous papers (Toczyłowski & Zoltowska, 2009; Zoltowska & Toczyłowski, 2008).

The individual uplift R_d^D covers the difference between the maximum possible benefit $\tilde{B}_d^D(\vec{\pi}_n)$ of buyer d located in node n and her actual benefit $B_d^D(\vec{\pi}_n, \hat{p}_d^D)$ obtained when realizing operational schedule $\{\hat{p}_d^D\}$ under hourly prices $\vec{\pi}_n$:

$$R_d^D \geq \tilde{B}_d^D(\vec{\pi}_n) - B_d^D(\vec{\pi}_n, \hat{p}_d^D), \quad \forall n, d \in N_n^D. \quad (9)$$

To provide the formulation of maximum and actual consumer's benefits, first we define the auxiliary nonnegative surplus λ_{dhi}^+ and deficit prices λ_{dhi}^- . The surplus and deficit prices λ_{dhi}^+ , λ_{dhi}^- are declared for

each hourly bid-block i of the buyer d , as functions of prices at hour h , following the findings of [Toczyłowski \(2002\)](#):

$$\lambda_{dhi}^+ - \lambda_{dhi}^- = e_{dhi} - \pi_{nh}, \quad \lambda_{dhi}^+, \lambda_{dhi}^- \geq 0 \quad \forall n, d \in N_n^D, h, i. \quad (10)$$

If the surplus price λ_{dhi}^+ is positive, the corresponding block of energy is competitive (should be accepted in linear market clearing). If the energy bid is non-competitive (should be rejected in linear market clearing), its surplus price is zero. Analogously, deficit prices are positive for non-competitive blocks, otherwise are zero.

These auxiliary prices are useful in providing the optimal benefits of individual buyers. Total maximum benefit of the consumer d is simply equal to the sum of surplus prices times the amount of the energy from corresponding bid block. This is true, if no restrictions or constraints were applied on the bids. However, when shifting constraints (1) are considered, the total benefit is restricted by the minimum $E_d^{\min}$ and maximum $E_d^{\max}$ amount of energy to be consumed.

- 1 First, let us explain the effect imposed by the maximum amount of energy $E_d^{\max}$. Consider a 2-hour 3-block example, as depicted in [Fig. 2](#). The hourly prices π_{nh} are reasonably low, however it is more profitable for buyer d located in node n to curtail the consumption in 1'st hour and shift it to 2'nd hour with the lower price. Notice, that the energy in 2'nd block of the bid in 1'st hour is curtailed, even though the benefit from consumption of this energy is grater than zero (the surplus price λ_{d12}^+ is grater than zero). Still the benefit from shifting is grater. Total maximum benefit is then a sum of highest partial benefits, restricted by the amount $E_d^{\max}$, as shown in [Fig. 2](#).
- 2 Next, the effect imposed by the minimum amount of energy $E_d^{\min}$ is illustrated. If $E_d^{\min} = E_d^{\max}$ in our example, and very high price in 1'st hour occurs, then consumer accepts some loss of a benefit. The energy in 2'nd block is curtailed to the maximum amount that can be shifted, the remaining part is consumed even though the benefit from consumption of this energy is less than zero (this means that the deficit price λ_{d12}^- is grater than zero). Total maximum benefit is then a sum of highest partial benefits, reduced by the sum of lowest partial losses, until the minimum amount $E_d^{\min}$ is reached, as shown in [Fig. 3](#).
- 3 Lastly, there are still possible situations, that the total benefit is simply equal to the sum of nonzero partial benefits. This is depicted in [Fig. 4](#), where no restriction on the $E_d^{\min}$ is submitted.

To determine the maximum benefit $\bar{B}_d^D$ of the consumer d we apply the findings of [Ogryczak & Tamir \(2003\)](#). The authors show that the value of Θ_k defined as the sum of k largest components of $\vec{u}$ in $\mathcal{R}^m$ may be found by solving the linear program:

$$\Theta_k = \min \left(ky + \sum_{j=1}^m s_j^+ \right), \quad (11)$$

$$s.t. \quad s_j^+ \geq u_j - y, \quad s_j^+ \geq 0, \quad j = 1 \dots m, \quad (12)$$

where y is a free variable representing the k th largest component of $\vec{u}$, while s_j^+ represents the surplus of each component of $\vec{u}$ over y ([Ogryczak & Tamir, 2003](#)).

In our case we are interested in some of the largest surplus prices λ_{dhi}^+ of consumer d , name y_d , together with some of its lowest deficit prices, that is largest minus deficit prices $-\lambda_{dhi}^-$, name g_d . From [Eq. \(10\)](#) follows that y_d is always non-negative, while g_d is always non-positive.

Each surplus price higher than y_d , denoted as s_{dhi}^+ constitutes the part of buyers maximum benefit, which is the product of the surplus s_{dhi}^+ and the amount of the energy from the corresponding bid block $\bar{p}_{dhi}^D$. Naturally, y_d also contributes to the buyers maximum benefit, multiplied by $E_d^{\max}$, [Figs. 2–4](#).

Similarly, g_d reduces the buyers maximum benefit, multiplied by $E_d^{\min}$. Each deficit price higher than g_d , denoted as z_{dhi}^+ contributes to the buyers maximum benefit, in the amount multiplied by the energy from the corresponding bid block $\bar{p}_{dhi}^D$, [Fig. 3](#). This observation yields the following mathematical model:

$$\bar{B}_d^D = E_d^{\max} y_d + E_d^{\min} g_d + \sum_{h,i} \bar{p}_{dhi}^D (s_{dhi}^+ + z_{dhi}^+), \quad \forall d, \quad (13)$$

$$s_{dhi}^+ \geq \lambda_{dhi}^+ - y_d, \quad s_{dhi}^+ \geq 0, \quad \forall d, h, i, \quad (14)$$

$$z_{dhi}^+ \geq -\lambda_{dhi}^- - g_d, \quad z_{dhi}^+ \geq 0, \quad \forall d, h, i. \quad (15)$$

Notice, that $\bar{B}_d^D$ is minimized through [Eq. \(9\)](#), as the uplifts are minimized, so [Eq. \(13\)](#) states proper realization of the maximum benefit.

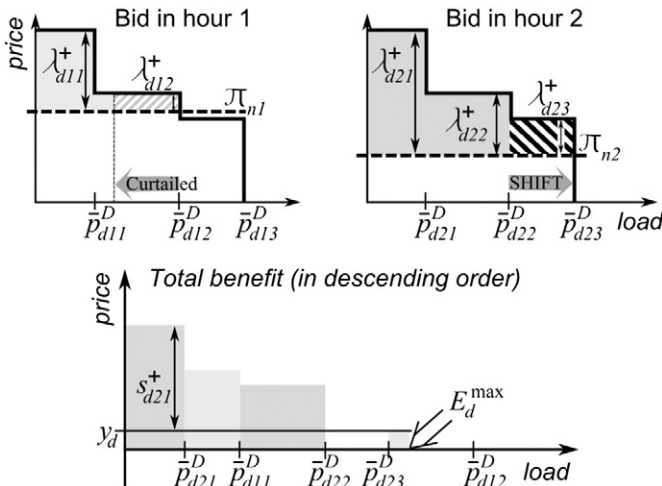


Fig. 2. Bids and cumulative profit under low market prices: maximum benefit is restricted because the maximum amount of energy to be consumed is restricted.

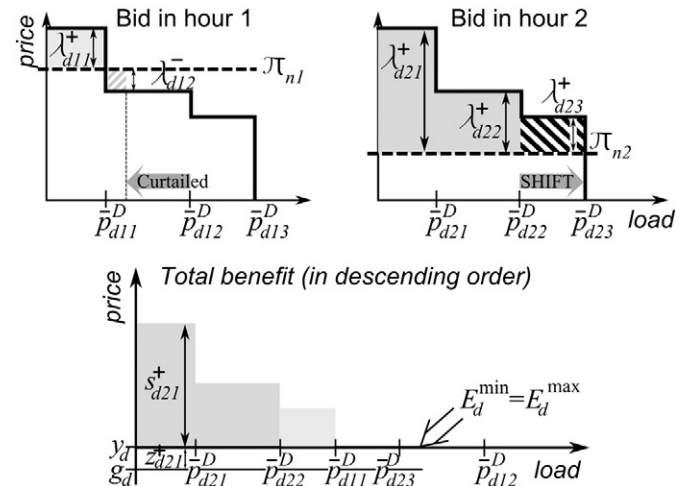


Fig. 3. Bids and cumulative profit under high market prices: maximum benefit is reduced because the minimum amount of energy constraint must be met.

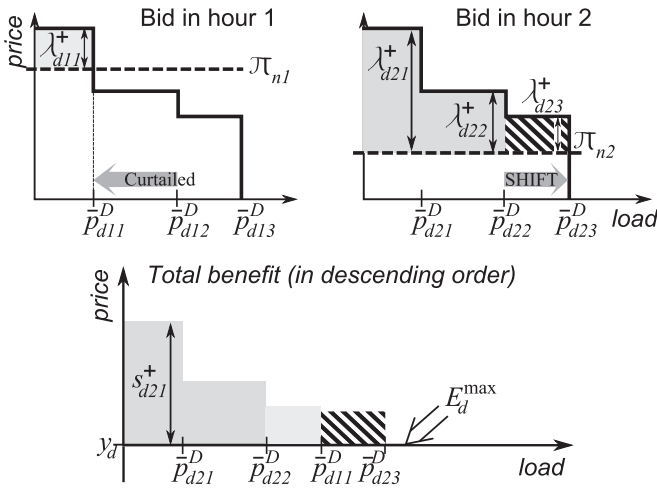


Fig. 4. Bids and cumulative profit: maximum benefit is simply equal to the sum of nonzero partial benefits.

Lastly, actual benefit function of each customer d have to be incorporated into the constraints, which is straightforward, as the operational schedules $\{\hat{p}_d^D\}$ are already determined:

$$B_d^D = \sum_{d,h,i} (e_{dhi} - \pi_h^D) \cdot \hat{p}_{dhi}^D, \quad \forall n, d \in N_n^D. \quad (16)$$

Notice, that the newly defined constraints indicating uplift and benefit of buyer with shifting requirement are formulated in the linear form.

4.2. Supply uplift and profit formulation

The MILP model of uplifts assigned to generators, who submit multipart offers was first developed by [Toczyłowski & Zoltowska \(2009\)](#). Further, the model was extended with various possible representations of generators' offers by [Zoltowska & Toczyłowski \(2008\)](#). Thus in this paper only the relevant constraints are included, for more detailed explanation the interested reader is referred to the previous papers.

Generally, the idea is to separately consider the dispatch $R_{lh}^{S,v} = 1$ and commitment-based uplifts $R_l^{S,UC}$, under the assumption that the dispatch at each period can be decoupled. The individual dispatch-related uplift $R_{lh}^{S,v} = 1$ at hour h covers the difference between the generator's l maximum possible profit $\tilde{P}_{lh}^S(\pi_{nh})$ and her actual profit $P_{lh}^S(\pi_{nh}, \hat{p}_{lh}^S)$ obtained when realizing operational schedules $\{\hat{p}_{lh}^S\}$ under price π_{nh} :

$$R_{lh}^{S,v=1} \geq \hat{v}_{lh} \cdot \tilde{P}_{lh}^S(\pi_{nh}) - P_{lh}^S(\pi_{nh}, \hat{p}_{lh}^S), \quad \forall h, n, l \in N_n^S. \quad (17)$$

The $R_{lh}^{S,v} = 1$ uplift covers only dispatch profit losses of generators committed at hour h (i.e. $\hat{v}_{lh} = 1$).

First, similarly to costumers we define the auxiliary nonnegative surplus λ_{lhi}^+ and deficit prices λ_{lhi}^- for each hourly block i of the generator l , as functions of prices at hour h ([Toczyłowski, 2002](#)):

$$\lambda_{lhi}^+ - \lambda_{lhi}^- = \pi_{nh} - c_{lhi}, \quad \lambda_{lhi}^+, \lambda_{lhi}^- \geq 0, \quad \forall h, n, l \in N_n^S, i. \quad (18)$$

The auxiliary prices are used to provide the optimal hourly profit $\tilde{P}_{lh}^S(\pi_{nh})$:

$$\tilde{P}_{lh}^S = \sum_i \lambda_{lhi}^+ \cdot \bar{p}_{lhi}^S - \lambda_{lhi}^- \cdot \bar{p}_{lhi}^S, \quad \forall l, h. \quad (19)$$

Notice, that the optimal hourly profit may be negative if a generator is dispatched at the minimum $\bar{p}_{lh1}^S$.

Actual hourly profit is simply given as:

$$P_{lh}^S = \sum_i (\pi_{nh} - c_{lhi}) \cdot \hat{p}_{lhi}^S, \quad \forall h, n, l \in N_n^S. \quad (20)$$

Following on, one must derive the individual commitment-related uplift $R_l^{S,UC}$ that covers the difference between the generator's l total maximum profit $\tilde{P}_l^{S,UC}(\bar{\pi}_n)$ and her actual total profit equal to sum of P_{lh}^S over h , minus commitment dependent costs UC_l . In this paper we assume only start-up costs SU_l , other instances of UC_l are investigated in previous paper ([Zoltowska & Toczyłowski, 2008](#)):

$$R_l^{S,UC} \geq \tilde{P}_l^{S,UC}(\bar{\pi}_n) - \sum_h P_{lh}^S - UC_l(\bar{v}_l), \quad \forall n, l \in N_n^S. \quad (21)$$

As [Toczyłowski & Zoltowska \(2009\)](#) have shown, optimal profit function $\tilde{P}_l^{S,UC}$ can be determined with the use of the Bellman inductive functions $\omega_{l,h+1}^{on}$ and $\omega_{l,h+1}^{off}$ denoting the optimum profits from hour 0 to $h+1$ for the l th generator in a state 'on' or 'off':

$$\omega_{lh}^{off} = \max\{\omega_{l,h-1}^{off}; \omega_{l,h-1}^{on}\}, \quad \forall l, h \in H/1 \quad (22)$$

$$\omega_{lh}^{on} = \tilde{P}_{lh}^S + \max\{\omega_{l,h-1}^{off} - SU_l; \omega_{l,h-1}^{on}\}, \quad \forall l, h \in H/1, \quad (23)$$

where

$$\omega_{l1}^{off} = 0, \quad \omega_{l1}^{on} = \tilde{P}_{l1}^S - SU_l, \quad \forall l. \quad (24)$$

In Eq. (24) we assume all the generators are uncommitted in the beginning of the market horizon.

To guarantee proper realization of the $\max\{\}$ functions, binary variables x_{lh}^{off} and x_{lh}^{on} are used along with sufficiently large parameters T_l :

$$0 \leq \omega_{l,h-1}^{off} \leq \omega_{lh}^{off} \leq \omega_{l,h-1}^{off} + (1 - x_{lh}^{off}) T_l, \quad (25)$$

$$\omega_{l,h-1}^{on} \leq \omega_{lh}^{off} \leq \omega_{l,h-1}^{on} + x_{lh}^{off} T_l, \quad (26)$$

$$\tilde{P}_{lh}^S + \omega_{l,h-1}^{on} \leq \omega_{lh}^{on} \leq \tilde{P}_{lh}^S + \omega_{l,h-1}^{on} + x_{lh}^{on} T_l, \quad (27)$$

$$0 \leq \omega_{lh}^{on} - \tilde{P}_{lh}^S - \omega_{l,h-1}^{off} + SU_l \leq (1 - x_{lh}^{on}) T_l, \quad (28)$$

where all the constraints (25)–(28) are defined for every $l \in S$ and every $h \in 2, \dots, |H|$.

Lastly, the greater value of $\omega_{l,|H|}$ constitutes the maximum overall price-dependent profit $\tilde{\omega}_l$:

$$\tilde{P}_l^{S,UC} \geq \omega_{l,|H|}^{off}, \quad \tilde{P}_l^{S,UC} \geq \omega_{l,|H|}^{on}, \quad \forall l. \quad (29)$$

4.3. Transmission constraints

Transmission constraints considered in the auction lead to varying the prices between the nodes, so as to reflect the fact, that a low price node is not able to supply power across a constrained line ([Hogan et al., 1996](#)).

To properly reflect the locational value of power we use exactly the same LMP decomposition formula of prices as defined in Eq. (8), wherein shadow prices are replaced with new variables to be computed:

$$\pi_{nh} = \pi_{mh} - \sum_k SF_k^n (\eta_{kh}^+ - \eta_{kh}^-), \quad \forall n, h, \quad (30)$$

where m is the reference node, and η_{kh}^+ , η_{kh}^- are the congestion prices of line k at hour h . The price η_{kh}^+ is positive only if line k is congested at hour h , while η_{kh}^- is positive if opposite side congestion occurs:

$$\eta_{kh}^+ (U_k - \hat{P}_{kh}) = 0, \quad \eta_{kh}^- (U_k + \hat{P}_{kh}) = 0, \quad \forall k, h. \quad (31)$$

Note that if there are no transmission constraints, the congestion prices are zero and prices at every node are identical. Otherwise, the congestion prices η may differ from the shadow prices γ and the surplus CR (congestion rent) resulting from prices differentiation may differ from the one obtained under LMPs:

$$CR = \sum_{n,h} \pi_{nh} \cdot \left(\sum_{k:k_2=n} \hat{P}_{kh} - \sum_{k:k_1=n} \hat{P}_{kh} \right). \quad (32)$$

Theoretically, a congestion price should reflect the reduction in variable cost that results from an increase by 1 MW in the capacity of the congested line. Still, non-convexities may hinder the shadow prices from this property. In this new approach the relative flexibility in determining congestion prices allows for adjusting the locational prices to values resulting in the lowest possible uplifts, while the network influence is preserved.

4.4. Objective: uplift vs. prices differentiation

Constraints defined in previous subsections, reflecting demand and supply uplift and benefit/profit as well as network constraints (9)–(10), (13)–(21), (24)–(31) are enough to state the complete, minimum-uplift pricing model with objective:

$$\text{Uplift} \equiv \min \sum_d R_d^D + \sum_l R_l^S, \quad (33)$$

where R_l^S is the sum of dispatch- and commitment-related uplift for unit l ($R_l^S = R_l^{S,UC} + \sum_h R_{lh}^{S,v} = 1$).

However, the previous pricing model by Toczyłowski & Zoltowska (2009) goes even further, that is, it includes also budget balance constraint. The idea is based on differentiation of buying and selling prices to collect the uplift, so no funds outside the auction are required. In the network unconstrained auction it can be proved that the difference (weighted with demand consumption) between buying and selling prices is equal exactly to minimum uplift (Toczyłowski, 2002). Here we shall show that this relation holds also in a transmission-constrained environment.

Let M be the buyers' payments and uplifts:

$$M = \sum_{n,h} \pi_{nh}^D \sum_{d \in N_n^D} \sum_i \hat{P}_{dhi}^D - \sum_d R_d^D, \quad (34)$$

where price π_{nh}^D is the buying price in node n , hour h .

Similarly, on the generation side let C be the sellers' charges:

$$C = \sum_{n,h} \pi_{nh}^S \sum_{l \in N_n^S} \sum_i \hat{P}_{lhi}^S + \sum_l R_l^S, \quad (35)$$

where price π_{nh}^S is the selling price in node n , hour h .

To get a balanced budget we require $M = C + CR$ (CR is congestion rent), so:

$$\sum_{n,h} \pi_{nh}^D \sum_{d \in N_n^D} \sum_i \hat{P}_{dhi}^D - \sum_d R_d^D = \sum_{n,h} \pi_{nh}^S \sum_{l \in N_n^S} \sum_i \hat{P}_{lhi}^S + \sum_l R_l^S + CR. \quad (36)$$

We can eliminate $\hat{P}_{lhi}^S$ from Eq. (36), as in each node n holds: $\sum_{l \in N_n^S} \sum_i \hat{P}_{lhi}^S - \sum_{d \in N_n^D} \sum_i \hat{P}_{dhi}^D = \sum_{k:k_1 \vee k_2 = n} \hat{P}_{kh}$. Then, after changing the order we can write:

$$\sum_{n,h} (\pi_{nh}^D - \pi_{nh}^S) \sum_{d \in N_n^D} \sum_i \hat{P}_{dhi}^D - \sum_{n,h} \pi_{nh}^S \sum_{k:k_1 \vee k_2 = n} \hat{P}_{kh} = R + CR. \quad (37)$$

The second term on the left hand side of Eq. (37) is simply the definition of congestion rent (32). Then the first term on the left hand side of Eq. (37) is equivalent with the overall uplift $R = \sum_d R_d^D + \sum_l R_l^S$.

We showed, that the minimum uplift is equivalent with the minimum differences between locational buying and selling prices weighted by the node's demand, when transmission constrained auction is priced, i.e.:

$$\text{Uplift} \equiv \min \sum_{n,h} (\pi_{nh}^D - \pi_{nh}^S) \sum_{d \in N_n^D} \sum_i \hat{P}_{dhi}^D. \quad (38)$$

4.5. Pricing model summary and solution procedure

General direct minimum-uplift pricing model (DMU) is formulated as a multi-period mixed-integer programming problem with convex objective (33) and linear constraints (9)–(10), (13)–(21), (24)–(31).

When the uplift costs cannot or should not be charged to fixed demands (for example when only elastic demands participate in the auction) then hourly locational buying and selling prices π_{nh}^D , π_{nh}^S ($\pi_{nh}^D \geq \pi_{nh}^S$) are calculated in the direct minimization of uplift (DMU) model, with objective (38) used instead of Eq. (33).

Furthermore, additional constraints (32), and (36) imposing budget balance must be included.

Finally, uniform market prices π_{nh} in constraints (9)–(10), (13)–(16) related to consumers uplift should be replaced with buying prices π_{nh}^D , while in constraints (17)–(21), (24)–(30) related to generators uplift and spatial price differentiation the uniform prices π_{nh} should be replaced with selling prices π_{nh}^S .

To apply the model the following procedure is proposed.

- 1 First, auction clearing problem (1)–(7) is solved, which provides optimal schedules to be priced;
- 2 LMPs are derived according to Eq. (8) from the relaxed solution to Eqs. (1)–(7) auction clearing problem with fixed commitment variables;
- 3 Restricted DMU pricing model is solved, with additional constraints (39) imposing fixed values of prices π_{nh} (or π_{nh}^S) set to LMPs:

$$\pi_{nh} = \text{LMP}_{nh}, \quad \forall n, h. \quad (39)$$

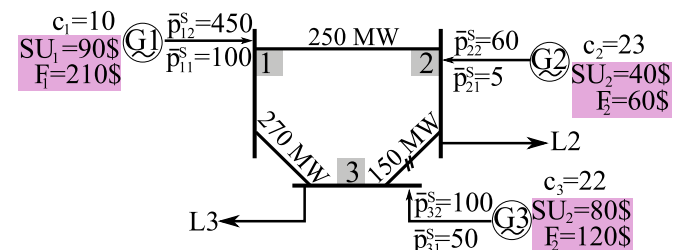


Fig. 5. Three-node example: system and offers.

Table 1
Three-node example: L3 bids' components [MW].

Bid	Hour	Case 0	Case 1	Case 2	Case 3
Fixed component	1	220	198	176	154
	2	489	440.1	391.2	342.3
Block size	1,2	0	14.8	28.36	42.54
$E^{min} = E^{max}$	1–2	0	70.9	141.8	212.7

This step allows us to compute a good initial feasible integer solution to the pricing problem. Also the loss and opportunity costs faced by consumers and generators under LMPs can be easily obtained from the solution of the restricted pricing problem.

- 4 Finally, the initial integer solution is communicated to CPLEX, so it is much easier to optimize the DMU pricing model in the uniform prices version or in buying/selling prices version.

Steps 1 and 2 constitute a typical pool-based auction mechanism. The two-step procedure given in steps 3 and 4 makes up the auction pricing. Apart from the LMP uplift, the solution from step 3 provides a good initial guess that enhances the computational performance of the original pricing model (Step 4).

5. Numerical results

The applications of the developed clearing and pricing models are studied on a simple 3-node example, and on the IEEE RTS 24-node 24-hour system.

5.1. Three-node example by Wu (2013)

We start with a basic three-node, two-hour example first studied by Wu (2013). The system data are depicted in Fig. 5. Three offers of generators are considered to supply demand in nodes 2 and 3. Two-block offers define energy at sale price c_i : first block states minimum level of generation $\bar{p}_{i1}^S$. Start-up costs SU_i (all generators are initially off) and fixed (no-load) costs F_i are also considered. Buyer L2 located in node 2 is inflexible and submits only the fixed part of hourly bids: 20 MW in hour 1 and 35 MW in hour 2. Buyer L3 located in node 3 may provide DR and submit elastic bids.

Following Wu (2013) we consider 4 levels of elasticity in L3 shifting bids:

Case 0: only fixed part of hourly bids: 220 MW in hour 1 and 489 MW in hour 2;

Case 1: 10% of load can be shifted;

Case 2: 20% of load can be shifted;

Case 3: 30% of load can be shifted;

Details of L3 bids are given in Table 1. The bids consists of a fixed part and 5 price-responsive blocks. Wu (2013) considers constant prices in each case in both hour: 20.8, 18.4, 16, 13.6, 11.2 \$/MWh for consecutive

Table 2
Case 0 results: dispatch [MWh], price [\$/MWh], payment, loss and uplift [\$].

Hour	Node	Dispatch		LMP		DMU	
		$\bar{p}_{dh}^D$	$\bar{p}_{th}^S$	Price	Loss	Price	Loss
1	1	–	240	10	300	10.47	98.00
	2	20	0	10	0	10.47	0
	3	220	0	10	0	10.47	0
2	1	–	420	10	210	10.67	20.00
	2	35	5	16	135	17.33	91.66
	3	489	99	22	200	24.00	16.67
L3 payment vs. uplift				12,958	845	22,027	226.33

Table 3

Case 1 results: dispatch [MWh], price [\$/MWh], payment, loss and uplift [\$].

Hour	Node	Dispatch		LMP		DMU	
		$\bar{p}_{dh}^D$	$\bar{p}_{th}^S$	Price	Loss	Price	Loss
1	1	–	288.9	10	300	10.47	75.18
	2	20.0	0	10	0	10.47	0
	3	268.9	0	10	0	10.47	0
2	1	–	422.5	10	210	10.67	18.33
	2	35.0	0	16	0	17.33	0
	3	440.1	52.6	22	200	24.00	94.80
L3 payment vs. uplift				12,371	710	13,377	188.31

blocks (in descending order). Only the corresponding load levels are different because of different levels of elasticity. All the curtailed load must be shifted to the other hour, so $E^{min} = E^{max}$ and equals the sum of load that can be shifted (i.e. in Case 1 $E^{min} = 0.1 \times (220 + 489) = 70.9$). Since only two hours are considered, then maximum elastic load in each hour equals E^{min} (i.e. in Case 1 $p_{31}^{max} = p_{32}^{max} = 5 \times 14.8 = E^{min}$).

All the clearing solutions to each case are the same as given by Wu (2013), even though our approach does not involve binary variables indicating the DR participation status. However, the prices vary, which is closely analyzed below. We compare our DMU prices with LMPs obtained by Wu (2013) and also with conventional LMPs derived according to Eq. (8) from the relaxed solution to Eqs. (1)–(7) auction clearing problem with fixed commitment variables.

5.1.1. Case 0: fixed demand bids

Table 2 shows the dispatch and pricing results when only fixed demand is considered.

LMPs calculated by Wu (2013) are identical with traditional LMPs – they ignore start-up and no-load costs and result in 845\$ loss faced by generators. New DMU prices are above LMPs, so they reduce the uplift by over 73%, but at the expense of increasing consumer fees, i.e. L3 pays about 63% more. This situation can be improved if demand elasticity is considered, which we show in next cases.

5.1.2. Case 1: 10% of L3 load may be shifted

Table 3 shows results for Case 1, that consider the possibility of shifting 10% of L3 load. All the admissible load is shifted, but the auction results does not differ significantly from the fixed demand case: LMPs and DMU prices remain unchanged, even though commitment of G2 is avoided (and so uplift is reduced). However, L3 payment is reduced by over 39% under DMU prices (the reduction effect under LMPs is less than 5% because the prices are lower).

5.1.3. Case 2: 20% of L3 load may be shifted

This case with higher volume of shiftable load shows very interesting properties of the newly proposed DMU pricing method. Load shifting capacity of L3 increases to 20% but LMPs fail in rewarding the demand flexibility: L3 payment is increased by over 30% comparing to

Table 4

Case 2 results: dispatch [MWh], price [\$/MWh], payment, loss and uplift [\$].

Hour	Node	Dispatch		LMP		DMU	
		$\bar{p}_{dh}^D$	$\bar{p}_{th}^S$	Price	Loss	Price	Loss
1	1	–	377.8	10	300	10.47	52.36
	2	20.0	0	10	0	10.47	0
	3	317.8	0	10	0	10.47	0
2	1	–	418.8	10	210	10.67	20.80
	2	35.0	7.4	23	100	17.33	141.94
	3	391.2	0	36	1200	24.00	0
L3 payment vs. uplift				17,261	610	12,715	215.10
Opportunity cost (G3)					1200		0

Table 5

Case 3 results: dispatch [MWh], price [\$/MWh], payment, loss and uplift [\$].

Hour	Node	Dispatch		LMP Wu (2013)		LMP		DMU	
		Energy	Who	Price	Loss	Price	Loss	Price	Loss
1	1	341.5	G1	10	300	10	300	10.47	50.63
	2	20.0	L2	10	0	10	0	10.47	0
	3	321.5	L3	10	0	10	0	10.47	0
2	1	422.5	G1	10	210	10	210	10.67	20.80
	2	35.00	L2	14.2	0	12.4	0	12.88	0
	3	387.5	L3	18.4	9.58	14.8	0	15.27	0
L3 payment vs. uplift				10,345	510	8 950	510	9 283	68.97
L3 benefit loss					9.58		0		0

Case 0 and 1 under LMPs (again, traditional LMPs are the same as computed by Wu (2013)). Uncommitted G3 generator cannot set LMP price in node 3 and faces large opportunity cost (i.e. $1200\$ = (36 - 22) \times 100 - (80 + 120)$), however it is traditionally not recovered so we mark this loss with emphasize in Table 4.

On the contrary, DMU prices remain the same as in previous cases, so the L3 payment continues to fall. DMU prices directly account for profit losses and opportunity costs, which allowed to avoid price spike in node 3, while network influence is preserved. The overall market condition is improved under DMU, with one exception to G2 profit loss, slightly higher than profit loss under LMP. Still, total profit loss (uplift) is over 64% lower than LMP uplift. Moreover no opportunity costs under DMU prices exist.

5.1.4. Case 3: 30% of L3 load may be shifted

L3 shifting capacity is increased to 30%, that is to 212.7 MW (66 in hour 1 plus 146.7 in hour 2), but due to the congestion only 101.5 MWh is shifted from hour 2 to 1. This fact should enable L3 to set the price in node 3 hour 2, however the straightforward approach adopted by Wu (2013) results in consumer benefit loss. Let us examine this situation more closely.

Wu (2013) examines the adequate prices independently in particular hours and claims the price in node 3 hour 2 to be equal to partially accepted shifting bid price, that is to $18.4 \$/\text{MWh}$ (second block of elastic bid is partially accepted as $342.3 + 42.54 + 2.66 = 387.5$) however one should notice that also elastic bid in hour 1 is partially accepted (fourth block of elastic bid with price 13.6 is partially accepted as $154 + 42.54 + 42.54 + 42.54 + 39.88 = 321.5$). With price equal to 10 in hour 1, in hour 2 price cannot be too high, as this creates higher motivation to shift load to first hour: possible unrealized benefit is equal to $(13.6 - 10) \times (42.54 - 39.88) = 9.576\$$ and so is the uplift required by L3.

On the contrary, traditionally calculated LMPs account for that case: price in node 3 hour 2 is 14.8 which creates benefit in hour 2 equal exactly to unrealized benefit in hour 1 ($(18.4 - 14.8) \times 2.66 = 9.576$). L3 achieves maximum benefit under traditional LMPs, no consumers uplift is needed.

Finally, DMU prices that account for both consumer and generators profits minimize also uplift required to cover generators loss: price in node 1 is slightly higher than LMP price and covers most of G1 start-up and no-load cost. Benefit of L3 is decreased in hour 1 under DMU (unrealized benefit is now equal to $(13.6 - 10.47) \times (42.54 - 39.88) = 8.326$) so higher price comparing to standard LMP is required in hour 2 ($(18.4 - 15.27) \times 2.66 = 8.326$). Still DMU price ($15.27 \$/\text{MWh}$) is lower than LMP ($18.4 \$/\text{MWh}$) suggested by Wu (2013) and lowest uplift is required (68.97 vs. 510\$).

5.1.5. Buying and selling prices differentiation

So far all the DMU results were investigated with the assumption that buying and selling prices are uniform and the uplift is collected outside the market, at the expense of fixed demands. However, when

we apply the approach based on the buying and selling prices differentiation, as suggested in this paper, the results are the same, namely two general cases appear: (1) buying price in fixed-demand node 2 is increased, so the uplift is collected at the expense of fixed demands; (2) selling price in load-only node (for example node 3 in Case 3) is decreased, so the uplift is collected at the expense of congestion rent.

The method proposed to allocate the uplift should be more relevant in auction with all buyers price-responsive.

5.2. IEEE RTS

More realistic 24-hour case study is based on IEEE RTS system (Grigg & Wong, 1999), which consists on 24 nodes, 32 generators and 17 consumers. Network characteristics, generators start-up cost and time (all generators are off, but available at the beginning of auction horizon), as well as constant 4-block offers follow the one hour case study provided by Garcia-Bertrand et al. (2006). First block in offer defines the minimum generation level.

Consumers are located in first 20 nodes (1–10, 13–16 and 18–20), with base (fixed) loads given in Table 5 by Grigg & Wong (1999). Hourly percentage demand changes are considered for winter load profiles with a peak of 2850 MW in hours 18 and 19 (Table 4 for 51st week Tuesday's hourly peak load (Grigg & Wong, 1999)), which define hourly fixed components FD_{dh} . Besides the fixed component each demand submits elastic 5-block bids. The prices are constant and equal: 14, 13, 12, 11, 10 \$/MWh for consecutive blocks (in descending order). The corresponding amount of energy offered to buy in each block depends on the case:

Fixed (F0): Base case with zero load in elastic bids, i.e. $\bar{p}_{dhi}^D = 0$;
Elastic (E10, E20, E30): Part of fixed component FD_{dh} is offered to buy through elastic bids. Different levels of demand response (elasticity) are analyzed: 10%, 20% and 30% of fixed load may be consumed through elastic bids, total load must not exceed original fixed demand, i.e. for 10% of demand response $E_d^{max} = 0.1 \cdot \sum_h FD_{dh}$. Energy quantities in each block of elastic bids are simply equal to the fifth of E_d^{max} , which means that all flexible load could be shifted to a single hour. Minimum elastic load is zero ($E_d^{min} = 0$) so the daily consumption may be reduced, by the amount not exceeding respectively 10, 20 or 30% of base load.
Shifted (S10, S20, S30): Same as Elastic case, but with assumption that all curtailed load must be shifted, i.e. $E_d^{min} = E_d^{max}$.

Table 6 shows the clearing results: generation cost decreases with higher demand response, however when shifting constraints are applied further increase of flexibility (above 20%) does not affect generation cost, only consumers' willingness to pay increases. In Elastic case the overall social welfare increases mainly due to load reduction – Fig. 6. It can be seen that the load profiles flatten in both cases Elastic and Shifted, but the average load is much higher in the Shifted case.

The solution to pricing problems was obtained following the procedure described in Section 4.5. Standard MILP solver of Cplex 12.5 on a desktop computer with a 2.4-GHz i7 processor and 8 GB of RAM found the optimal solution in time not exceeding 70s in each case.

Table 6RTS clearing results: offered cost and willingness to pay in $10^3\$$.

Case	F0	E10	E20	E30	S10	S20	S30
Offered cost	513.89	459.15	412.54	376.68	494.99	493.19	493.19
Willingness to pay	0	52.39	106.18	154.54	79.48	158.96	238.44
Number of 'on' units	22	22	20	17	22	20	20
Load: min.	1681.5	2034.5	2107.4	2066.7	2207.7	2255.8	2277.5
max (MW)	2850.0	2565.0	2354.1	2178.7	2565.0	2488.3	2451.0

Both versions of DMU model were analyzed, i.e. uniform as well as buying/selling prices differentiation version. Table 7 summarizes the collective results.

As expected demand response (DR) payments increase with the elasticity in Shifted cases, as more energy is bought through price-responsive bids. But also in Elastic cases DR payments increase, which means that it is profitable to buy energy in originally low load periods.

In Elastic cases the consumers' payments, as well as uplifts and congestion rent decrease progressively with elasticity increase. Peak load, and the overall load is reduced so it is easier to fulfill the network and generation constraints, however the number of committed units is also continuing to fall.

In Shifted cases the load profiles are quite close to each other, regardless of elasticity level. What is counter-intuitive is the consumers payments, as well as uplifts and congestion rent increase in case S30, while the generation cost remains the same as in S20 case. The explanation is the solution multiplicity – different schedules were obtained in the cases discussed, and the price is completely flat (see Fig. 7); indeed Cplex reported 3 optimal solution saved. But outcomes are dramatically different for those social welfare-indifferent schedules. This problem with unit commitment models was pointed out by many researches (e.g., Chattopadhyay & Momoh, 1999).

Uplifts are not significant, comparing to total generation cost, however new DMU prices allow to achieve even better results in all cases, but E10 and S10. Notice, that the number of committed units is the same as in Fixed case, but prices decrease reflecting demand shift – see Fig. 7, which means that losses of units 1 and 6 committed only in high load periods (F0, E1, S1 cases) cannot be easily avoided. Fig. 7 shows prices in node 1, where the unit 1 is located.

The effect of uplift minimization is obtained with a rather small deviation from LMPs. When buying and selling prices differentiation is

Table 7

RTS pricing results: uplifts in \$, consumers payments, demand response (DR) payments and congestion rent (CR) in 10³\$ under LMPs and uniform as well buyer/seller differentiated DMU prices.

	Method	F0	E10	E20	E30	S10	S20	S30
Uplift	LMP	422	744	342	0	586	83	250
	DMU	49	522	29	0	523	28	135
Cons. paym.	LMP	968.72	862.10	741.75	645.58	953.10	796.05	929.81
	DMU	969.09	878.12	745.72	645.58	962.90	797.70	935.44
	π^D/π^S	969.04	876.82	745.68	645.58	960.3	797.92	936.91
	π^D/π^S	0	43.09	88.82	128.78	86.92	173.06	278.94
DR paym.	LMP	0	43.09	88.82	128.78	86.92	173.06	278.94
	DMU	0	43.46	88.65	128.78	86.95	173.46	280.63
	π^D/π^S	0	43.39	88.63	128.78	86.92	173.45	281.05
	π^D/π^S	0	43.39	88.63	128.78	86.92	173.45	281.05
CR	LMP	209.30	147.65	112.89	72.76	186.57	154.97	180.85
	DMU	212.25	156.07	117.67	72.76	190.44	154.91	188.38
	π^D/π^S	212.25	156.27	117.29	72.76	189.02	154.91	187.93
	π^D/π^S	212.25	156.27	117.29	72.76	189.02	154.91	187.93

examined we can observe that they are close to uniform DMU prices, and to LMPs naturally. Fig. 8 shows the prices in S10 case, when payments differ more significantly, regarding on the pricing method applied. It is interesting to observe however, that the consumers payments (for fixed demand as well as for demand responsive load) are slightly lower, comparing to the uniform DMU results. Still, the lowest payments are achieved under LMPs, even with the uplift cost included.

In the cases analyzed all consumers' uplift was always equal zero.

6. Conclusion

The challenge for competitive market is to include the demand bids in such a way, that the most efficient market dynamics are achieved. This paper gives the answer how the demand participants with shifting requirements can be included in the pool-based auction with non-

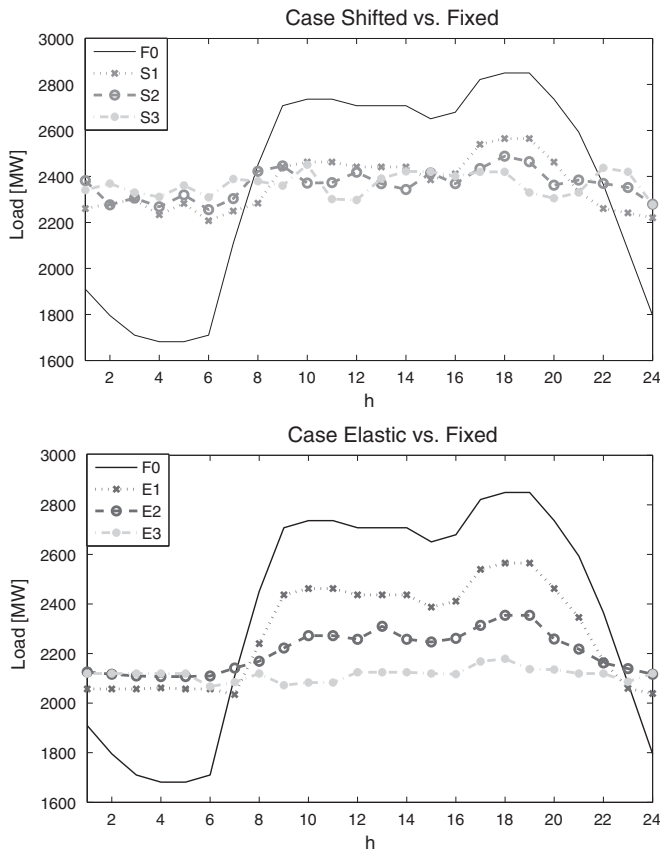


Fig. 6. RTS loads in cases Fixed, Elastic and Fixed.

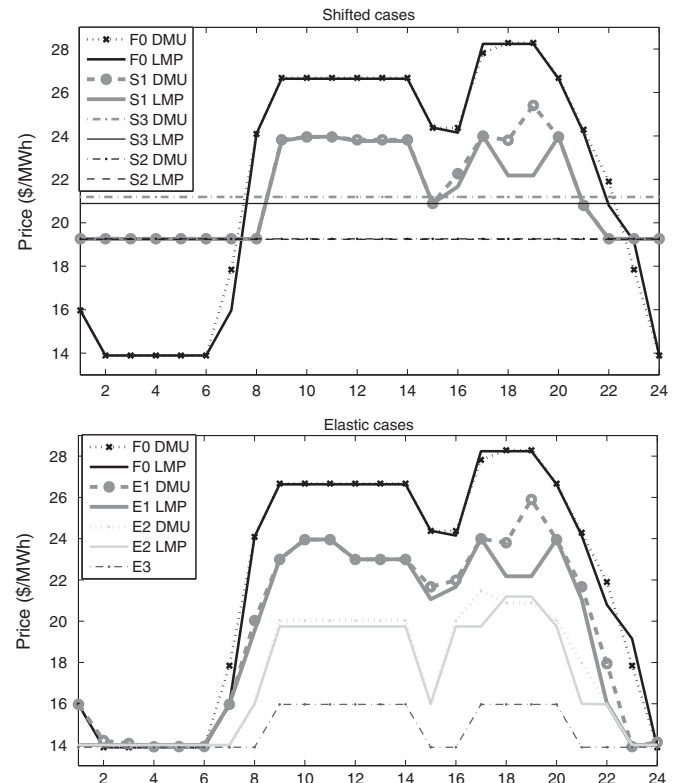


Fig. 7. RTS hourly prices in node 1.

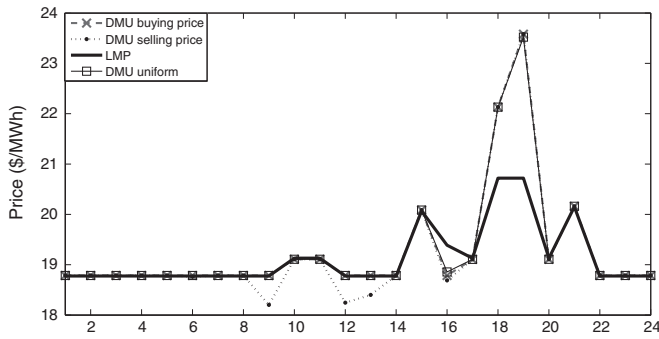


Fig. 8. RTS S10 hourly prices in node 13 under different pricing approaches.

convexities. The shifting bids developed protect the customers against over-curtailment, while market prices allow them to sustain the maximum individual benefits.

The numerical examples show that the complex market with non-convexities and transmission constraints may benefit from the inclusion of the active demands with shifting bids. The amount of unfairly priced generators will tend to decrease, without reducing the total amount of generation. Still, the pricing method must be carefully chosen, as the uplift reduction may lead to increased consumers payments. On the other hand, the traditional LMPs proved to be efficient in pricing the shifting demands.

Nonetheless, numerous criteria should be applied to judge any auction mechanism: individual rationality, consumers payments, budget balance, incentive compatibility, to name just a few most significant. Within that, further studies are required to justify the newly proposed clearing and pricing models. The preliminary findings reported in this paper can challenge the research to continued explore of prices abilities to properly signal the cost of supply to price responsive demands; as well as of the influences between network congestion and non-convexities.

The scheme developed may be used for example in future smart grid employing the double auctions to support the energy exchange between the micro-grid elements creating the system.

Acknowledgment

This research was partly supported by the Polish National Budget Funds 2010–2013 under grant no. N N514 044438.

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Nomenclature

A. Indices

- $d \in D, N_n^D$: Buyers (demands), subset located in node n
- $h \in H$: Periods of the auction horizon
- $i \in I$: Blocks of stepwise bid/offer
- $k \in K$: Transmission lines, with starting nodes $k1$ and ending nodes $k2$
- $l \in S, N_n^S$: Sellers (generators), subset located in node n
- $n \in N$: Network nodes

B. Parameters

- c_{lth} : Unit price of energy offered to sell in block i of l th sale offer in hour h
- $E_{dh}^{min}, E_{dh}^{max}$: Minimum/maximum daily buyer's d elastic load
- e_{dhi} : Unit price of energy offered to buy in block i of d th elastic bid in hour h
- FD_{dh} : Fixed demand of buyer d in hour h
- $\bar{P}_{dhi}^D$: Maximum amount of energy offered to buy in block i of d th elastic bid in hour h
- $\bar{P}_{dh}^{max}, \bar{P}_{dh}$: Maximum/expected elastic load in hour h
- $\bar{P}_{lth}^S$: Maximum amount of energy offered to sell in block i of l th sale offer in hour h ; $\bar{P}_{lth1}^S$ is the minimum level of generation
- SF_k^n : Shift factor of line k with respects to node n
- UC_i : Commitment-dependent costs, such as fixed cost F_i , start-up cost SU_i etc. in l th offer
- U_k : Capacity limit of line k
- v_i : Operating constraints submitted in l th offer

C. Variables in auction clearing problem

- P_{dhi}^D : Energy consumed in block i of d th elastic bid in hour h
- P_{kh} : Power flow in hour h in line connecting k
- P_{lth}^S : Energy sold in block i of l th sale offer in hour h
- v_{lh} : On/off status of l th generator in hour h
- λ_h : LMP price in hour h in reference node; dual variable of power balance constraint (10)
- $\gamma_{kh}^+, \gamma_{kh}^-$: Shadow price of congestion for flow/counterflow in line k , hour h ; dual variable of constraints (7)

D. Solution of auction clearing used in pricing problem

$\hat{p}_{dhi}^D$:	Optimal amount of energy consumed in block i of d th elastic bid in hour h
$\hat{p}_{gh}^D$:	Optimal power flow in line k , hour h
$\hat{p}_{lhi}^D$:	Optimal amount of energy sold in block i of l th sale offer in hour h
$\hat{v}_{lh}$:	Optimal status of l th generator, hour h

E. Variables in pricing problem

$\bar{B}_d^D, B_d^D$:	Maximum/actual benefit of buyer d
CR:	Congestion rent
η_{kb}^+, η^{-ab} :	Nonnegative congestion price of congestion/countercongestion in line k , hour h
g_d :	Lowest j th deficit price λ_{dhi}^- from all $i \in I$ blocks in all $h \in H$ hours of buyer d , where $j = E_d^{\min}$.
$\lambda_{dhi}^+, \lambda_{dhi}^-$:	Non-negative surplus/deficit of bid price e_{dhi} to buying price in hour h

$\lambda_{dhi}^+, \lambda_{dhi}^-$:	Non-negative surplus/deficit of selling price to offer price c_{lhi} in hour h
$\omega_{lh}^{off}, \omega_{lh}^{on}$:	Maximum profit from hour 0 to h for l th seller when in state 'off'/'on' in hour h
$\pi_{nlb}, \pi_{nlb}^D, \pi_{nlb}^S$:	Price in node n , hour h , buying/selling price
$\hat{p}_{lh}^D, \hat{p}_{lh}^S$:	Maximum/actual profit of l th generator in hour h
$\hat{p}_l^{S,UC}$:	Total maximum profit of seller l
R_d^D :	Uplift (compensation) paid to buyer d
R_l^D :	Total uplift (compensation) paid to generator l
$R_l^{S,UC}$:	Commitment-related uplift, paid to generator l
$R_{lh}^{S,v} = 1$:	Dispatch-related uplift, paid to generator l committed in hour h
S_{dhi}^+ :	Non-negative surplus of λ_{dhi}^+ over y_d
$x_{lh}^{off}, x_{lh}^{on}$:	Binary variable used to properly define $\omega_{lh}^{off}, \omega_{lh}^{on}$
y_d :	Largest j th surplus price λ_{dhi}^+ from all $i \in I$ blocks in all $h \in H$ hours of buyer d , where $j = E_d^{\max}$
z_{dhi}^+ :	Non-negative surplus of g_d over λ_{dhi}^-